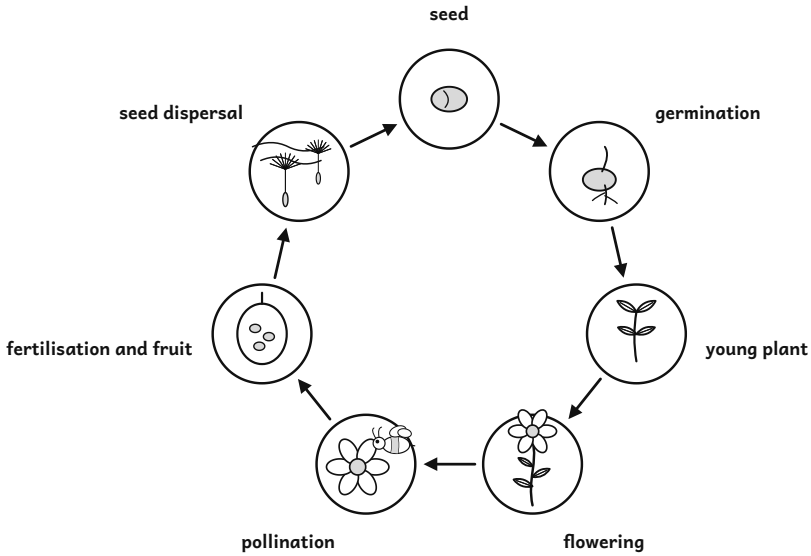
LIVING THINGS

# Plant Growth and Development

Flowering and non-flowering plants, the life cycle of a flowering plant and measuring growth




**In this pack**

- **Learn it:** How plants grow ★☆☆
- **Fact check:** Plant life cycle ★☆☆
- **Think it through:** Growing and comparing ★★☆
- **Investigate:** How fast does a bean plant grow? ★★☆
- **Real-world case study:** A rice farmer's year ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 1 Plants (1.1–1.4).

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · A2 · PLANT GROWTH

# How plants grow

★☆☆ Grade 3

Name: Class: Date:

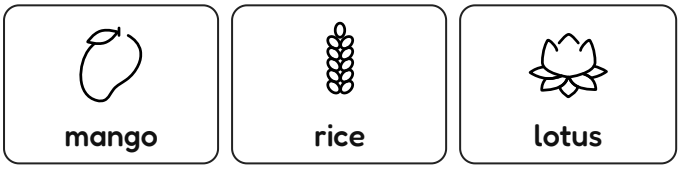
**Learning goals**

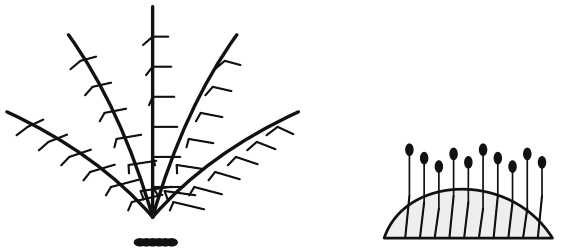
- Tell flowering and non-flowering plants apart.
- Describe the life cycle of a flowering plant and its main parts.
- Measure plant growth and compare plant and animal life cycles.

A tiny seed can grow into a giant mango tree. How does it happen? Let's follow a plant through its life.

## Flowering and non-flowering plants

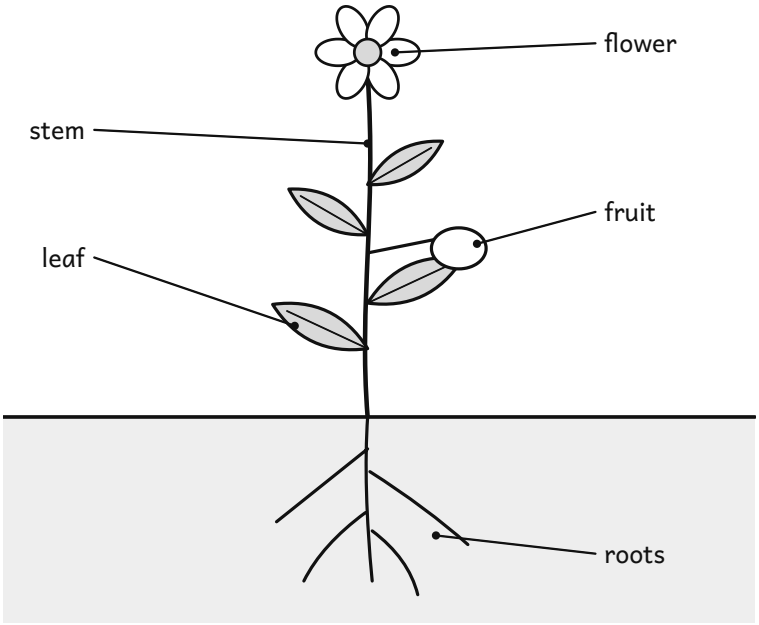
**Flowering plants** grow flowers that make seeds: mango, rice, lotus, sunflower, grass. **Non-flowering plants** do not have flowers: **ferns** and **mosses** grow from tiny **spores**, and pine trees make seeds in **cones**.



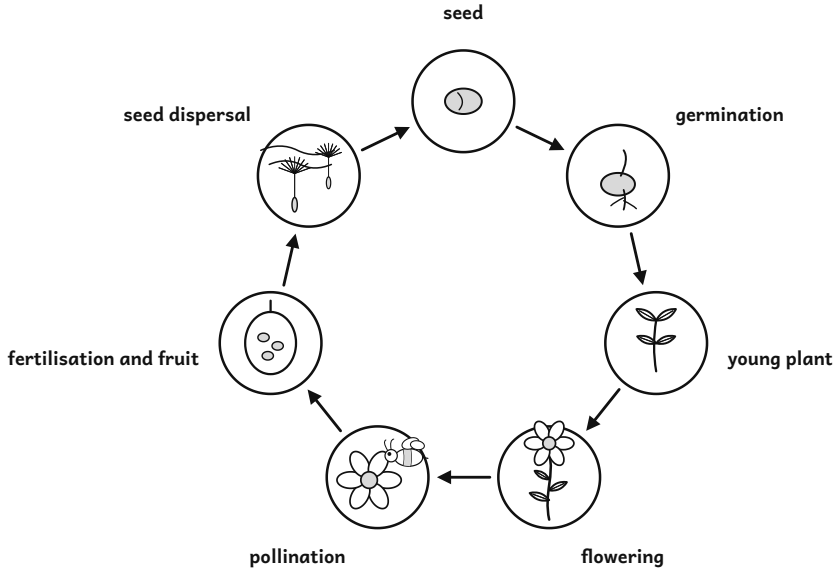

*Ferns and mosses: no flowers.*

## The parts of a flowering plant




*Root, stem, leaf, flower and fruit.*

## The life cycle of a flowering plant




*From seed to seed.*

- A **seed germinates** (starts to grow) when it has water and warmth.
- It becomes a **seedling** with its first leaves.
- The seedling grows into a **mature** (adult) plant.
- The plant makes **flowers**. Insects or wind carry pollen, and **seeds** form inside **fruits**.
- The seeds are spread, and the cycle starts again.

## Measuring growth

We can measure a plant's **height** with a ruler each week, count its **leaves**, or measure the width of its stem. A bar chart shows how it grows over time.

## Plants and animals

| | Flowering plant | Chicken |
|---|---|---|
| starts as | seed | egg |
| young | seedling | chick |
| adult | plant with flowers | hen |
| new life from | seeds | eggs |

**Bangladesh connection**

Rice is the most important crop in Bangladesh. Farmers plant rice seedlings in flooded fields and harvest the grains (seeds) a few months later.

**Key words**

**germinate** start to grow from a seed
**seedling** a young plant
**mature** fully grown
**spore** a tiny part that grows into a fern or moss
**life cycle** the stages of life, from seed to seed

FACT CHECK · A2 · PLANT GROWTH

# Plant life cycle

★☆☆
Grade 3

Name: Class: Date:

**Learning goals**

- Sort plants.
- Order the life cycle.

## 1 Sort the plants. [6]

SORT THESE

**rice** **mango tree** **fern** **pine tree** **sunflower** **moss**

| Flowering | Non-flowering |
|---|---|
| | |
| | |
| | |

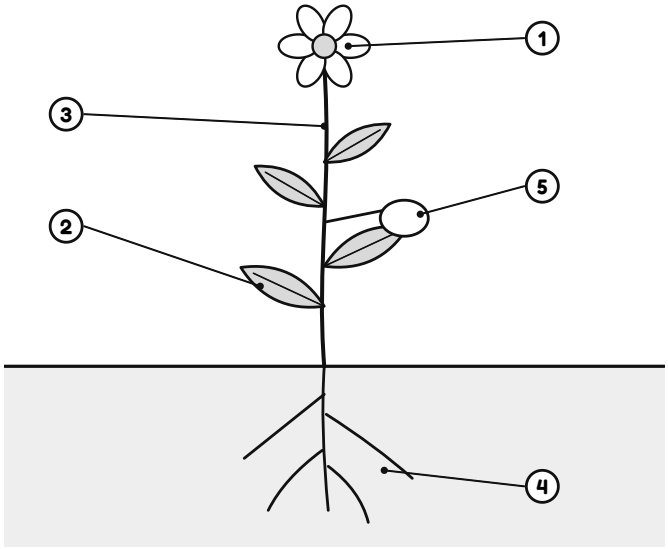
## 2 Number the stages in order. [4]

- ☐ fruit with new seeds
- ☐ seed
- ☐ adult plant with flowers
- ☐ seed germinates
- ☐ seedling

**3** **Label the plant.** [5]

LABELS

**flower leaf roots stem fruit**




1 _____ 2 _____ 3 _____

4 _____ 5 _____

THINK IT THROUGH · A2 · PLANT GROWTH

# Growing and comparing

★★☆
Grade 3

Name: Class: Date:

**Learning goals**

- Compare life cycles.
- Explain plant needs.

**1** **How is a plant life cycle like a chicken's? How is it different?** [2]

**2** **A seed is kept dry in a packet. Will it germinate? Why?** [2]

**3** **Draw the life cycle of a bean plant with four stages and arrows.** [3]

Bean life cycle

INVESTIGATE · A2 · PLANT GROWTH

# How fast does a bean plant grow?

★★☆ Grade 3

**Name:** **Class:** **Date:**

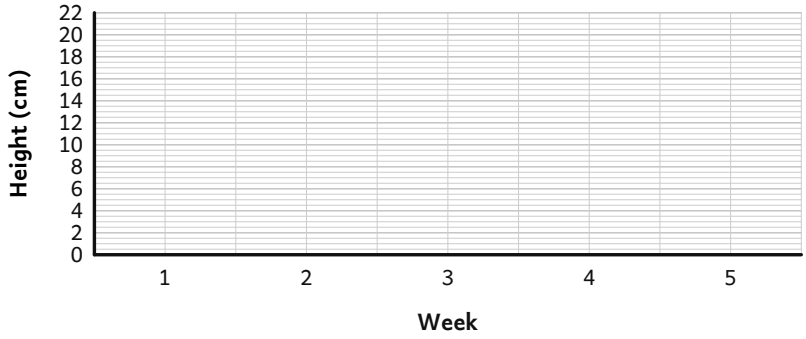
**Learning goals**

- Measure height each week.
- Draw a bar chart.
- Find a pattern.

Class 3C measured a bean plant every week.

**1** **Draw a bar chart.** [3]

| Week | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Height (cm) | 2 | 6 | 11 | 17 | 20 |




**2** **Describe the pattern.** [1]

.....

.....

**3** **In which week did it grow the most?** [1]

.....

**4** **Predict the height in week 6.** [1]

.....

REAL-WORLD CASE STUDY · A2 · PLANT GROWTH

# A rice farmer's year

★★★ Grade 3

**Name:** **Class:** **Date:**

**Learning goals**

- Follow a real crop through its life cycle.

Science | Farming | Maths

Abdul is a rice farmer in Mymensingh. First he soaks rice seeds in water until they **germinate**. He plants them close together in a small bed of mud. After about a month the **seedlings** are moved to the big flooded field.

The rice plants grow tall and green. Tiny **flowers** appear, and the wind carries pollen. The grains (**seeds**) grow and turn golden. About four months after planting, Abdul **harvests** the rice. He keeps some seeds to plant next year.

Based on rice farming in Bangladesh.

**1** **Put Abdul's year in order.** [4]

- ☐ harvest the grains
- ☐ soak seeds so they germinate
- ☐ rice flowers
- ☐ grow seedlings in a mud bed
- ☐ move seedlings to the field

**2** **Why does Abdul keep some seeds?** [1]

.....

**3** **Rice takes about 4 months from planting to harvest. How many weeks is that?** [1]

.....

# Show what you know

★★★
Grade 3

Name: Class: Date:

**Learning goals**

- Use plant words.

## 1 Find the words.

| C | P | N | E | G | R | O | W | L |
|---|---|---|---|---|---|---|---|---|
| R | N | F | E | R | N | K | B | P |
| W | O | N | A | R | R | O | O | T |
| J | M | N | F | L | O | W | E | R |
| C | S | G | S | T | E | M | O | R |
| F | E | L | K | S | S | P | S | V |
| C | E | E | I | W | E | E | K | I |
| V | D | A | K | V | Y | H | L | O |
| I | B | F | T | F | R | U | I | T |

**fern flower fruit leaf root seed stem**

## 2 Which plant has no flowers? [1]

- ◯ mango
- ◯ fern
- ◯ rice

## 3 A seed starting to grow is called... [1]

- ◯ germinating
- ◯ flowering
- ◯ wilting

## 4 Challenge. Why do plants make so many seeds? [2]

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| sort flowering and non-flowering plants | ○ | ○ | ○ |
| name the parts of a plant | ○ | ○ | ○ |
| draw a plant life cycle | ○ | ○ | ○ |
| measure plant growth | ○ | ○ | ○ |
| compare plant and animal life cycles | ○ | ○ | ○ |